Course Name:	Large Language Models and Query Engineering
Course Number:	0135444
Project Name:	LLM Application with Retrieval and Tools
Project Grade:	**30 Marks**

1. Project Description and Objectives

Teams choose a real problem, ideally one that involves Arabic text, and build an application that answers questions from documents it retrieves and calls at least one tool, then measure how well it works.

Objectives:

- Apply prompting, retrieval-augmented generation, tool use and evaluation from the course to one working system.
- Build and document a dataset and a test set for a real need.
- Evaluate the system for answer quality, faithfulness to sources, tool-call success, cost and safety, in Arabic where the project is Arabic.

Expected result: a working application with citations, a measured evaluation and a red-team report, presented in a live demo.

2. Project Requirements

- Teams of 3–4 students, registered on Moodle by the week 4 lab; every member writes code.
- Open-weight models only (e.g. Qwen2.5-1.5B, 3B or 7B Instruct) and a multilingual embedding model; no paid API.
- Everything runs in the department's GPU lab or on a free Colab T4.
- Public or properly licensed documents only, with source and licence recorded; no personal data about real people.
- AI assistants are allowed for brainstorming, debugging and editing, with a short disclosure in each submission.
- The final report ends with a contribution statement signed by every member.
- All deliverables are submitted on Moodle by the start of that week's lab session; late work loses 10% per day, up to three days.

3. Project Design Requirements

- Retrieval over the team's own document collection (at least 50 documents or 100 pages), with answers that cite the passages used.

- At least one tool the model calls when a question needs it, e.g. a SQL database, a calculator or a public API.
- A test set of at least 40 questions with reference answers: at least 10 that need the tool, at least 5 the documents cannot answer, and at least half in Arabic for an Arabic project.
- Evaluation of answer correctness, faithfulness, citation accuracy, tool-call success, latency and token cost; an LLM judge must be checked against human judgement on at least 20 answers.
- A red-team log of at least 10 attempts in English and Arabic, including prompt injection hidden in a document.
- A repository that another team can run from its README.

4. Suggested Project Structure

1. Choose a problem and users; write ten questions they would ask.
2. Collect and clean the documents; record source and licence.
3. Write the test set and the datasheet.
4. Build retrieval (chunking, embedding, vector search) and the tool.
5. Connect generation with citations and tool calls; demonstrate the prototype.
6. Evaluate on the full test set, red-team the system and fix what you can.
7. Write the report and present the final demo.

5. Partial Deliverables

Phase 1 – Problem Definition and Planning (week 7, 10 marks)

- A 2-page proposal: problem and users, ten example questions (tool questions included), documents and licence, the tool, risks, and team roles.
- Deliverable: proposal PDF on Moodle.

Phase 2 – Design and Implementation (week 10, 10 marks)

- The cleaned document collection, the test set (40+ questions with references), a one-page datasheet and a one-page system design (pipeline diagram and first model and chunking choices).
- Deliverables: dataset files, test set, datasheet and design PDF on Moodle.

Phase 3 – Testing, Evaluation and Final Submission (weeks 12–15, 10 marks)

- Week 12: working prototype demonstrated in the lab, with a repository that runs from its README (2 marks).
- Week 14: evaluation on the full test set and the red-team log, 3–4 pages (3 marks).
- Week 15: 10-minute live demo with questions, final report (at most 8 pages) and tagged code (5 marks).

Submission: electronic copies only on Moodle; reports as PDF; code as a repository link or zip.

6. Scope and Final Outcome

Project Requirements Specification Form – Teaching and Examination Process Monitoring Procedures / Faculty of Science and Information Technology	QF01/2650-4.0

Team project: a question-answering application over the team's own documents that retrieves, cites and calls at least one tool, built on open-weight models within the lab's compute. Complete when the prototype runs from its README, the full test set and red-team log are reported, and the final demo, report and code are submitted in week 15. Arabic-language projects are preferred.